

BRINTIK MAJUMDER

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Professional Summary

FULL-STACK AI ENGINEER | SYSTEMS ARCHITECT | DECOUPLED ML/UI SPECIALIST

Highly analytical full-stack AI engineer specializing in **system thinking** and the architecture of decoupled, high-performance applications. Expertise lies in connecting heavy **machine learning backends (YOLOv8, PyTorch)** via **low-latency APIs (FastAPI)** to deliver premium, **Awwwards-tier frontend interfaces (React, Framer Motion)**. Proven **Automation-First Engineering Approach** to build complex systems that unify disparate data sources, streamline technical workflows, and translate complex business problems into scalable, production-ready AI solutions.

Tech Stack

- **AI/ML:** Python, YOLOv8, PyTorch, Groq API, LLaMA 3.1
- **Frontend:** React, Vite, Framer Motion, Tailwind CSS
- **Backend/DevOps:** FastAPI, Docker, Vercel, Git/GitHub

Featured Portfolio Projects

AI Image Recognizer & Analytics Platform

- **Challenge:** Develop a low-latency, scalable computer vision platform capable of real-time object detection and contextual spatial analytics.
- **Solution:** Architected a decoupled system using **YOLOv8** for the ML backend and exposed its inference logic via a high-throughput **FastAPI** service. Built a **React/Vite** frontend utilizing low-latency API integration to display live bounding boxes and structured data analysis, including **relative percentage mapping** for spatial component assessment.
- **Result:** Achieved end-to-end inference and visualization latency under 100 ms, providing near real-time operational feedback.

Interactive Portfolio Engine (LLM-Grounded UI)

- **Challenge:** Create a visually premium, interactive portfolio experience that uses Generative AI to provide personalized, context-aware user engagement.
- **Solution:** Designed an Awwwards-tier UI/UX using **React** and **Framer Motion** for micro-interactions and smooth transitions. Integrated the **LLaMA 3.1** model via the **Groq API** to enable real-time conversational response grounded in project data, implementing **LLM context-tracking** to maintain dialogue state and coherence over multi-turn interactions.
- **Result:** Elevated portfolio engagement with a fluid, personalized experience, leveraging Groq's high-speed inference for instant, low-latency LLM responses.

Professional Experience

Indus Net Technologies

Consulting Business Representative | Kolkata, India | May 2025 – July 2025

- Architected automated reporting pipelines, unifying disparate data sources and leveraging Power BI for dynamic visualization.
- Translated complex business requirements into comprehensive technical specifications and actionable engineering roadmaps.
- Created critical process documentation, bridging business objectives with technical implementation teams.

ITPeopleNetwork

Jr. Business Development Executive | Kolkata, India | August 2024 – April 2025

- Engineered solutions to streamline data flow and reporting efficiency by integrating CRM systems with backend APIs.
- Architected a Power BI-based automated reporting system to rapidly process data and identify key business trends.

Education

University of Calcutta

MBA in Marketing and Operations | Kolkata, WB, India | August 2022 – July 2024

Siliguri Institute of Technology

B. Tech in Electronics and Communication Engineering | Siliguri, WB, India | August 2017 – July 2021